lab08_Goes

```
import os
import polars as pl

base_dir = r"\\smb\ra266580\Documentos\ME315"

basics_path = os.path.join(base_dir, "title.basics0.tsv.gz")
ratings_path = os.path.join(base_dir, "title.ratings.tsv.gz")
principals_path = os.path.join(base_dir, "title.principals0.tsv.gz")

basics_lazy = pl.scan_csv(basics_path, separator="\t", null_values=r"\N").select(["tconst", "
ratings_lazy = pl.scan_csv(ratings_path, separator="\t", null_values=r"\N").select(["tconst"
principals_lazy = pl.scan_csv(principals_path, separator="\t", null_values=r"\N").select(["t
```

1

```
q1 = (
  basics_lazy.filter(pl.col("titleType") == "movie")
  .join(ratings_lazy, on="tconst")
  .sort(["averageRating", "numVotes"], descending=[True, True])
  .select(["primaryTitle", "averageRating", "numVotes"])
  .limit(5)
  .collect()
)

print(q1)
```

```
shape: (5, 3)

 primaryTitle              averageRating   numVotes
```

```
 ---                         ---             ---
 str                         f64             i64

 Kaveri                      10.0            1023
 Kurukku                     10.0            451
 Jedal Dar Omghe 30 Metri    10.0            142
 Sargashte                   10.0            134
 Gorgeous Rascal             10.0            115
```

2

```
q2 = (
    basics_lazy.filter(pl.col("titleType") == "movie")
    .join(ratings_lazy, on="tconst")
    .filter(pl.col("averageRating") > 8)
    .with_columns(pl.col("genres").str.split(","))
    .explode("genres")
    .filter(pl.col("genres").is_not_null())
    .group_by("genres")
    .len()
    .sort("len", descending=True)
    .limit(1)
    .collect()
)

print(q2)
```

```
shape: (1, 2)

 genres        len
 ---           ---
 str           u32

 Documentary   10564
```

3

```
principals_atores = principals_lazy.filter(
    pl.col("category").is_in(["actor", "actress"])
)

q3 = (
    basics_lazy.filter(pl.col("titleType") == "movie")
    .join(ratings_lazy, on="tconst")
    .filter(pl.col("averageRating") > 7.5)
    .join(principals_atores, on="tconst")
    .group_by("nconst")
    .len()
    .sort("len", descending=True)
    .limit(3)
    .collect()
)

print(q3)
```

```
shape: (3, 2)

 nconst      len
 ---         ---
 str         u32

 nm0004660   231
 nm0595934   155
 nm3183374   124
```

Os atores são , respectivamente: Dr. Rajkumar, Mohan Babu e T. N. Balakrishna.